

Where in Europe was this photo taken?

Vaibhav Prajapati

University of Technology Nuremberg (UTN) Nuremberg, Germany; **Email:** vaibhav.prajapati@utn.de

Abstract—Given a single street-level photograph, the task is to predict the coordinate it was taken at: a regression problem over the surface of the Earth. The model built here predicts that coordinate as a gated mixture of geo-cell centroids at two spatial resolutions, trained with distance-weighted cell targets and learned loss balancing. Arriving at it took a sweep over candidate backbones, a set of ablations that change one thing at a time across the architecture and the objective, and a study that removes one augmentation at a time. Almost none of that search moved the result. The two changes that did move it cost no parameters at all: training at the source images’ native resolution, and decoding the coordinate by nearest-neighbour search over the backbone’s own features instead of reading it off the trained heads. The two decoders also pull against each other, since augmentation teaches invariances while nearest-neighbour matching depends on the surface detail those invariances discard. What error remains is bimodal, hinging on whether the country is identified rather than on precision within it, and the retrieval result itself has to be read against how densely the photograph collection already covers the same ground.

I. METHOD

The task is to regress a latitude and a longitude from one image. The large vision–language backbones behind recent geolocation work [1]–[3] are out of reach at this size.

Data. The labelled set is 11,758 street-level photographs at 512×512 from twelve European countries, each with a latitude, a longitude and a country. Counts are nearly balanced, nine countries contributing 1,000 images and Iceland 960, Poland 934, Belarus 864, but coverage inside a country is not: the photographs are sampled along roads and lie in dense chains rather than spread over the map, which Sec. III returns to. Another 2,400 are held back unlabelled, and the metric is median great-circle error.

Architecture (Fig. 1). The model is one path with four readouts hanging off a single feature vector. A 512×512 image first goes through a small residual filter, two 3×3 convolutions through 16 channels and back to 3, whose output is added to the image. The result is then bilinearly resized to the backbone’s input size. That filter exists because of how GeoGuessr is actually played [4]: much of the signal sits in small detail, road signs, licence plates, shop lettering, the script a street name is written in, and a fixed downsample to a backbone’s usual 224px destroys exactly that. Making the downsample learned [5] was meant to keep what mattered. The resolution sweep then removed the problem instead: training settled at native 512px, where the resize is the identity, so the filter survives as a pre-filter on the full image. The backbone θ (HGNetV2-B0 [6], via `timm` [7]) reduces the image to a single 2048-dimensional feature vector f by averaging over its final feature map, and everything after that reads f and nothing else.

Two classification heads share one 2048→128 stem, literally the same module under two names, so each head sees an independently dropped view of one representation and the stem is paid for once. They produce distributions $p^{(64)}$ and $p^{(256)}$ over two k -means partitions of the training coordinates, fitted on the unit sphere rather than in degrees since otherwise cells get stretched by latitude

[8]. Each partition stores its cell centres as a fixed table, $C^{(64)}$ and $C^{(256)}$, not updated by training. Picking a single cell would quantize the prediction at the cell scale, so the expectation over cells is taken instead, recovering position inside a cell. A gate $g = \sigma(w^\top f)$, one linear unit on f , decides per image how far to trust the finer partition:

$$a = g p^{(256)\top} C^{(256)} + (1 - g) p^{(64)\top} C^{(64)}. \quad (1)$$

A fourth head reads f through its own 2048→128→2 projection and adds an offset δ , so the prediction is $(\hat{\varphi}, \hat{\lambda}) = a + \delta$. A weighted average of centroids can only ever land somewhere between them, so the blend alone can never predict outside the area they span; δ is what frees it. One rule runs through all of this: every added part starts as the identity. The filter’s last convolution, the offset head’s last layer and the gate are zero-initialised, so an untrained model is exactly the plain centroid blend. The gate starts at 0.5 and settles at 0.86–0.88, on a schedule no constant reproduces, low while the 256-way head is noisy and rising as it sharpens; fixed values were swept and none matched it.

That is 4,518,887 learnable parameters, or 4,531,841 counting the unlearned tensors too, 640 floats of cell centres and 12,314 normalisation statistics.

Objective. Cell targets decay with great-circle distance rather than being one-hot [3], [9], $t_i = \text{softmax}(-d_{\text{km}}(y, c_i)/\tau)$, $\tau = 25$ km, so confusing an adjacent cell costs less than confusing a distant one, which ordinary cross-entropy cannot express. The three losses are balanced by weights the model learns rather than weights set by hand [10], $L = \sum_i e^{-\rho_i} L_i + \rho_i$. The two choices are coupled: that weight is stationary at $1/L$ and one-hot cross-entropy reaches $\sim 10^{-4}$, which would send it to infinity, while soft targets floor the loss at the entropy of t .

Decoding rules. *Direct* is the model’s own output, $(\hat{\varphi}, \hat{\lambda}) = a + \delta$, used unchanged. *Country-snap* adds one step. Every cell sits in a known country, so summing cell probabilities by country gives a distribution over the twelve, the two partitions again weighted by g . The most likely country is kept, cells outside it are zeroed, the rest renormalised, and the anchor recomputed from those alone. *Retrieval* throws the heads away and runs in two passes once training is over. The index is built with the trained model itself, the same weights with the heads unused: every labelled image is embedded once and stored with its coordinate, and a query then takes its nearest entry’s coordinates under cosine similarity [11], [12]. Deployment is three stages, train, build the index, answer a query, and only the first learns anything. All three rules read one checkpoint and add no parameters, so any difference between them is decoding, not learning.

Protocol. Every configuration is run under both five-fold country-stratified cross-validation, which covers each image once, and a three-seed hold-out split (8,230/1,764/1,764). Selection uses hold-out, because cross-validation does not rank two configurations reliably when they sit within about 10 km of each other.

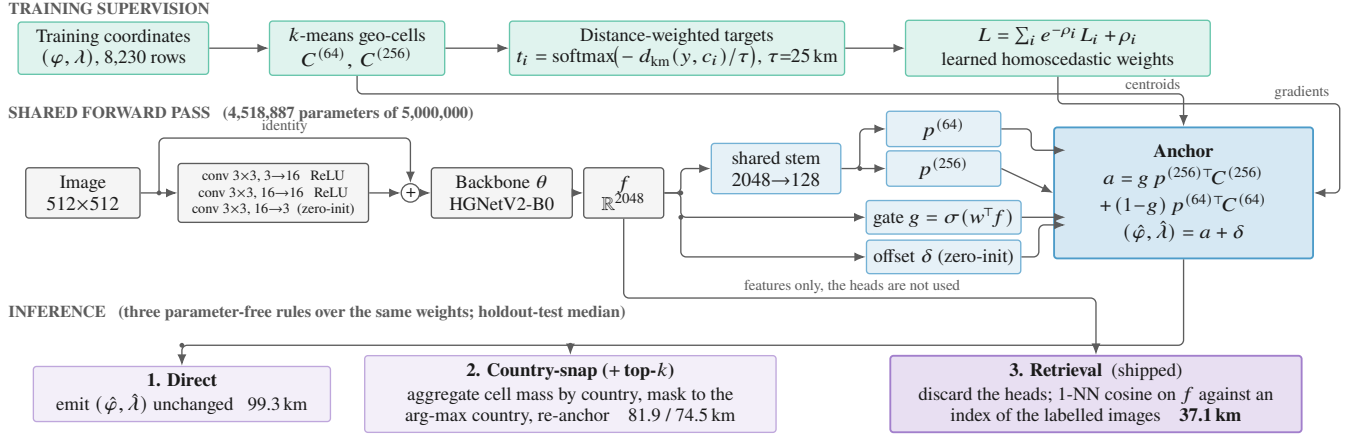


Fig. 1. The model. One backbone is trained once (middle band); the three decoding rules (bottom band) read those same weights and add no parameters. Only the forward pass carries parameters.

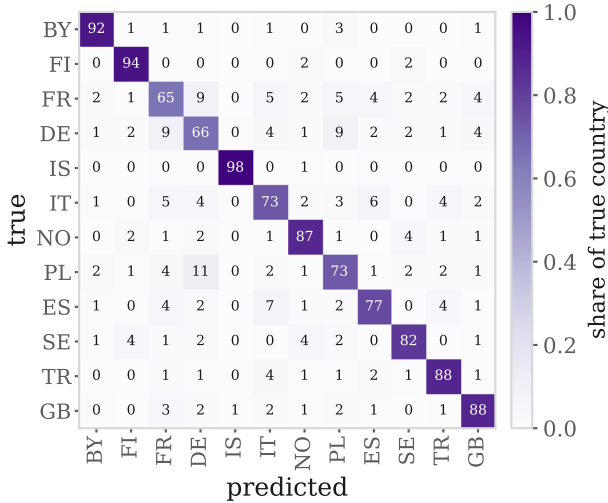


Fig. 2. Country confusion under retrieval, the best of the three decoding rules, row-normalised over the 11,758 cross-validation predictions. Each cell is the percentage of that row’s true country sent to that column, rounded to the nearest point.

II. RESULTS

Backbone. The sweep covers 25 backbones, eight runs each (Table A2). A plain CNN trained from scratch manages only 358.1 km, worse than the 288.1 km from ignoring the image and predicting the country centroid, so pretrained features [13], [14] do almost all of the work. The decoding columns also disagree: ConvNeXt-Femto-OLS wins every direct column, HGNetV2-B0 every retrieval one.

The pre-registered criterion, direct-prediction median, picks ConvNeXt-Femto-OLS. HGNetV2-B0 was adopted instead on retrieval, the rule that ships, worth 18.1 km on test and 18.7 km on cross-validation; Sec. III is what that costs. HGNetV2-B0 and EfficientViT-B0 are otherwise indistinguishable.

Ablations. (Tables A3, A4) change one thing each. Soft targets, the dual head with its gate and the offset head all earn their parameters; 64/256 is not improved by 32/128 or 128/512; the learned loss weights are neutral.

Input resolution. Every run from 288 to 512 px beats 256, and the trend never turns over before the source pixels run out. Training

error stays flat while the gap to validation error closes from 97 to 86 km, so the pixels help the model transfer rather than fit.

Augmentation. Switching the augmentation off costs 8.2 km on the direct head and *gains* 4.5 km under retrieval, because augmentation teaches invariances while retrieval wins by matching near-duplicates on the very detail those invariances throw away. No single arm accounts for it, so it stays and only the change that held up was taken: dropping random-resized-crop, worth 10.9 km on the direct path.

Final configuration. HGNetV2-B0 at native resolution, augmentation without resized crop, decoded by retrieval. Three seeds give 41.2 ± 1.2 km on hold-out test and 39.1 km on cross-validation; the deployed checkpoint reaches 37.1 km.

III. CONCLUSION

The error is sharply bimodal. When the retrieved neighbour lands in the right country the median error is 16.6 km; when it does not, it is 1,142 km, and 99.6% of country errors sit above the overall median. The mistakes are not spread evenly either: France and Germany are each other’s main confusion, and Poland is most often taken for Germany (Fig. 2). What the decoding rules do *not* do is fix the country: accuracy is 81.7%, 81.7% and 81.8% under the three, and country-snap cannot change it at all, snapping to the country the heads already chose. What separates them is precision once the country is right, the median falling from 59.6 to 54.5 to 16.6 km as the country-wrong cases worsen from 887 to 1,076 and 1,142 km.

What the retrieval number is really measuring. Retrieval does well partly because it only has to recognise a place it has already seen: splitting road-sampled chains at random within a country separates photographs, not places, and 28.7% of held-out images sit within a kilometre of a training image, the near-duplication that inflates benchmarks elsewhere [15]. The evaluation images have fewer such matches, so the real error should exceed 41 km.

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APPENDIX: COMPLETE RESULT TABLES

Table A1 gives the settings and the machine behind every number in this report. The three result grids follow: Table A2 the backbone sweep, Table A3 the architecture and input ablations, Table A4 the augmentation study. Conventions are shared. All figures are median great-circle error in km, lower is better. The hold-out validation, cross-validation and hold-out-test retrieval columns pool every run’s predictions together and take one median over the lot (5,292, 11,758 and 5,292 rows); **hold-out-test direct is the median of the three seed medians**, a different estimator, and must not be differenced against the others. In Tables A3 and A4 each median carries the dispersion $\pm$ of the 5 fold or 3 seed medians behind it. With $n \leq 5$ that is a descriptive spread, not a standard error, and no cell is a significance test. Much of those two tables lies inside its own noise, so rows should be read against their own row 1 rather than against each other. *gap* is the difference between validation and training error at the selected epoch, under cross-validation. Bold marks the best value per column.

TABLE A1

EVERYTHING NEEDED TO REPRODUCE THE REPORTED RUNS. VALUES ARE READ FROM THE SHIPPED CONFIG AND THE RUN LOGS, NOT RESTATED FROM MEMORY.

<i>Data and splits</i>	
Labelled images	11,758 at 512×512
Hold-out split	8,230 / 1,764 / 1,764, country-stratified
Cross-validation	5-fold, country-stratified
Seeds	42, 7, 123; geo-cell seed fixed at 42
<i>Model</i>	
Backbone	hgnetv2_b0, ImageNet-1k weights
Input / trunk resolution	512 / 512 px (resize is the identity)
Geo-cells	64 coarse, 256 fine, k -means on the sphere
Target bandwidth τ	25 km
Dropout	0.3
Regression loss scale	100 km
Loss-weight init λ	1.0
Top- k for the snap fallback	3
<i>Optimisation</i>	
Optimiser	AdamW [18], lr 3×10^{-4} , decay 10^{-3}
Schedule	cosine annealing [19], $T_{\max} = 100$
Batch size	64, four dataloader workers
Gradient clipping	5.0 (max norm)
Epochs	≤ 100 , early stop patience 10, min delta 0.5 km
Precision	full fp32 , no mixed precision
<i>Augmentation (train only)</i>	
Geometry	resize to 512; no random-resized-crop
Rotation	$\pm 15^\circ$
Perspective	distortion 0.2, $p = 0.5$
Colour jitter	brightness / contrast / saturation 0.2
Random erasing [20]	$p = 0.25$, scale 0.02–0.1
Horizontal flip	none (only the UK drives on the left)
Normalisation	ImageNet mean/std
<i>Hardware and software</i>	
GPU	1× NVIDIA A40, peak 17.5 GiB used
Throughput	~73 s/epoch
Wall clock	1.2–1.9 h per run (58–92 epochs)
Stack	Python 3.12, PyTorch 2.11 (CUDA 12.8), <code>time</code>
	1.0.28

TABLE A2

BACKBONE SWEEP. THE 24 ARMS THAT TRAINED, EACH UNDER BOTH PROTOCOLS (5 CV FOLDS + 3 HOLD-OUT SEEDS = 8 RUNS PER ARM) AT 256 PX ON THE FROZEN ARCHITECTURE, THEN SCORED UNDER ALL THREE DECODING RULES. SORTED BY HOLD-OUT TEST RETRIEVAL, THE RULE THAT SHIPS, WHICH IS DELIBERATELY *NOT* THE SELECTION CRITERION.

Backbone	Hold-out validation			5-fold cross-validation			Hold-out test		Params
	direct	snap	retr.	direct	snap	retr.	direct	retr.	
HGNetV2-B0 (adopted)	121.10	105.02	54.10	108.89	94.38	47.08	130.29	54.94	4,518,887
EfficientViT-B0	128.52	107.74	53.92	119.25	99.29	50.45	135.92	57.74	2,502,969
MobileNetV4-Conv-Small	146.75	128.50	69.94	133.48	112.76	59.80	158.50	66.64	2,866,985
InceptionNeXt-Atto	111.91	99.07	71.76	100.58	90.69	64.01	122.51	72.90	3,487,241
ConvNeXt-Femto-OLS	105.97	98.28	73.21	99.88	92.47	65.76	113.93	73.07	4,985,209
ConvNeXt-Femto	118.62	105.56	78.72	106.78	99.08	72.05	115.55	77.22	4,976,473
PVTv2-B0	126.52	113.78	84.10	125.46	112.39	84.05	131.45	80.09	3,520,553
MixNet-M	136.70	120.58	82.73	126.81	113.67	80.53	137.17	84.77	3,917,135
ConvNeXt-Atto	127.53	116.78	83.08	122.43	109.36	80.79	123.42	84.97	3,501,761
EfficientNet-B0	127.09	110.90	87.93	114.03	103.03	71.28	140.47	85.29	4,381,509
FastViT-T8	123.90	113.77	94.06	117.37	105.58	88.81	128.98	92.98	3,499,609
MobileOne-S0	170.12	149.38	89.30	150.94	135.14	81.85	187.94	95.44	4,576,441
ConvNeXt-Atto-OLS	157.25	139.16	89.19	144.83	131.31	86.72	164.25	97.33	3,509,153
EfficientFormerV2-S0	145.02	132.34	97.05	137.90	124.34	90.06	167.23	100.01	3,336,489
RepGhostNet-0.5x	195.69	166.95	99.71	189.43	163.80	84.10	211.74	106.27	1,406,929
GhostNet-1.0x	157.18	140.60	100.38	144.23	129.43	86.95	169.12	106.30	4,275,469
StarNet-S1	138.59	124.58	102.87	136.06	124.16	104.16	149.01	110.24	2,770,513
SwiftFormer-XS	151.82	133.62	108.15	141.94	128.74	93.85	159.05	111.36	3,134,901
RepViT-M1	146.41	128.73	117.36	135.99	123.82	111.84	156.72	130.98	4,857,401
CS3-DarkNet-Focus-S	186.74	164.48	130.86	177.51	154.51	121.99	184.41	133.55	2,938,313
EdgeNeXt-XX-Small	202.68	178.36	137.20	176.88	155.55	115.62	217.45	136.56	1,246,393
EfficientViT-M2	176.36	166.38	175.29	182.72	171.42	180.21	207.28	195.66	4,064,675
MobileViT-XXS	245.29	232.47	234.55	230.00	212.54	216.98	249.05	251.05	1,078,265
Plain CNN (no pretrained backbone)	344.33	307.62	229.08	325.14	288.93	212.74	358.13	274.12	3,503,369

TABLE A3

ARCHITECTURE AND INPUT ABLATIONS ON THE LOCKED BACKBONE, EACH ROW A SINGLE CONFIG DIFF FROM ROW 1.

Arm	Hold-out validation			5-fold cross-validation			Hold-out test		gap (km)
	direct	snap	retr.	direct	snap	retr.	direct	retr.	
reference HGNetV2-B0, 256 px	121.10 $\pm$ 11.5	105.02 $\pm$ 8.2	54.10 $\pm$ 3.8	108.89 $\pm$ 4.0	94.38 $\pm$ 4.8	47.08 $\pm$ 3.3	130.29 $\pm$ 13.0	54.94 $\pm$ 6.4	96
one-hot targets ($\tau=0$)	117.61 $\pm$ 9.5	103.34 $\pm$ 9.2	57.14 $\pm$ 2.2	106.79 $\pm$ 13.3	96.04 $\pm$ 11.5	52.71 $\pm$ 5.8	132.87 $\pm$ 17.7	55.67 $\pm$ 8.0	90
$\tau = 50$ km	126.65 $\pm$ 5.3	113.46 $\pm$ 9.4	53.44 $\pm$ 0.7	121.11 $\pm$ 6.1	108.06 $\pm$ 5.7	52.32 $\pm$ 5.7	139.45 $\pm$ 4.3	55.76 $\pm$ 4.2	102
single head, no gate	130.83 $\pm$ 3.1	119.23 $\pm$ 3.4	57.38 $\pm$ 3.7	126.18 $\pm$ 6.8	116.04 $\pm$ 7.4	55.22 $\pm$ 5.3	137.00 $\pm$ 6.7	58.18 $\pm$ 1.6	109
no offset head	125.40 $\pm$ 2.5	105.28 $\pm$ 3.2	54.37 $\pm$ 2.5	114.13 $\pm$ 3.8	98.11 $\pm$ 6.4	49.87 $\pm$ 5.1	137.18 $\pm$ 12.4	58.91 $\pm$ 8.2	89
fixed loss weights	118.16 $\pm$ 1.7	101.40 $\pm$ 0.1	54.51 $\pm$ 3.1	113.18 $\pm$ 9.5	96.45 $\pm$ 8.9	47.24 $\pm$ 5.9	119.43 $\pm$ 7.8	53.18 $\pm$ 5.8	98
no residual pre-filter	105.31 $\pm$ 4.3	91.83 $\pm$ 1.7	45.93 $\pm$ 1.8	103.12 $\pm$ 8.4	90.19 $\pm$ 7.3	39.16 $\pm$ 3.1	111.82 $\pm$ 2.1	45.31 $\pm$ 3.2	85
cells 32/128	122.42 $\pm$ 0.5	114.30 $\pm$ 3.5	60.33 $\pm$ 1.8	113.21 $\pm$ 3.2	103.57 $\pm$ 3.5	51.11 $\pm$ 5.2	128.74 $\pm$ 2.2	63.63 $\pm$ 1.9	95
cells 128/512	121.20 $\pm$ 3.1	99.09 $\pm$ 1.4	54.18 $\pm$ 2.9	107.26 $\pm$ 4.2	91.72 $\pm$ 2.9	47.08 $\pm$ 3.9	132.85 $\pm$ 10.2	54.23 $\pm$ 1.8	102
input 288 px	117.51 $\pm$ 11.6	101.44 $\pm$ 8.2	54.00 $\pm$ 3.2	101.80 $\pm$ 2.4	89.55 $\pm$ 3.4	42.84 $\pm$ 5.6	126.99 $\pm$ 9.7	54.13 $\pm$ 4.0	92
input 320 px	109.79 $\pm$ 4.2	96.46 $\pm$ 3.8	51.52 $\pm$ 4.5	105.57 $\pm$ 6.3	94.02 $\pm$ 4.8	42.79 $\pm$ 3.0	119.04 $\pm$ 10.4	52.01 $\pm$ 1.6	92
input 352 px	107.79 $\pm$ 6.5	94.64 $\pm$ 6.9	47.43 $\pm$ 1.2	100.43 $\pm$ 5.8	87.78 $\pm$ 6.2	40.88 $\pm$ 3.3	113.70 $\pm$ 4.8	46.68 $\pm$ 2.6	88
input 384 px	105.18 $\pm$ 3.0	93.49 $\pm$ 3.0	46.32 $\pm$ 2.2	99.21 $\pm$ 7.8	86.93 $\pm$ 6.0	41.37 $\pm$ 2.9	115.74 $\pm$ 3.6	45.00 $\pm$ 4.0	90
input 448 px	109.96 $\pm$ 7.5	96.14 $\pm$ 8.7	47.46 $\pm$ 4.4	98.53 $\pm$ 6.3	85.66 $\pm$ 5.7	38.48 $\pm$ 3.6	113.11 $\pm$ 8.0	47.33 $\pm$ 4.0	85
input 512 px (adopted)	104.58 $\pm$ 2.9	90.12 $\pm$ 1.0	45.95 $\pm$ 5.7	100.79 $\pm$ 5.1	88.29 $\pm$ 5.0	39.09 $\pm$ 2.2	108.97 $\pm$ 2.9	43.83 $\pm$ 3.7	86

TABLE A4

AUGMENTATION STUDY. LEAVE-ONE-OUT OVER THE FIVE FACTORS OF THE RECIPE, PLUS THE NO-AUGMENTATION FLOOR AND THE ONE FACTOR THE RECIPE DELIBERATELY EXCLUDES (HORIZONTAL FLIP, SINCE ONLY THE UK DRIVES ON THE LEFT). **THE BASELINE IS ROW 1, THE 512 PX MODEL OF TABLE A3,** NOT THE 256 PX REFERENCE: BENCHMARKING AGAINST THE EARLIER STAGE WOULD CREDIT THE RESOLUTION LOCK TO AUGMENTATION.

Arm	Hold-out validation			5-fold cross-validation			Hold-out test		gap (km)
	direct	snap	retr.	direct	snap	retr.	direct	retr.	
baseline HGNetV2-B0, 512 px	104.58 $\pm$ 2.9	90.12 $\pm$ 1.0	45.95 $\pm$ 5.7	100.79 $\pm$ 5.1	88.29 $\pm$ 5.0	39.09 $\pm$ 2.2	108.97 $\pm$ 2.9	43.83 $\pm$ 3.7	86
no augmentation at all	108.56 $\pm$ 1.9	94.30 $\pm$ 2.6	40.99 $\pm$ 0.5	103.64 $\pm$ 6.3	90.16 $\pm$ 3.3	37.14 $\pm$ 2.1	117.20 $\pm$ 1.6	39.37 $\pm$ 1.3	93
– random-resized-crop (adopted)	93.65 $\pm$ 4.3	83.48 $\pm$ 2.1	40.10 $\pm$ 1.7	92.81 $\pm$ 4.8	83.38 $\pm$ 4.7	39.10 $\pm$ 4.2	98.07 $\pm$ 3.1	41.22 $\pm$ 1.2	79
– rotation	108.64 $\pm$ 7.5	98.35 $\pm$ 7.3	47.80 $\pm$ 4.8	99.29 $\pm$ 3.8	88.97 $\pm$ 3.7	40.32 $\pm$ 2.8	112.44 $\pm$ 12.3	43.45 $\pm$ 5.1	88
– perspective	112.15 $\pm$ 3.5	96.89 $\pm$ 0.8	46.32 $\pm$ 3.9	105.93 $\pm$ 9.0	92.47 $\pm$ 7.3	40.72 $\pm$ 2.4	115.89 $\pm$ 5.9	41.35 $\pm$ 2.4	88
– colour jitter	104.73 $\pm$ 11.9	91.14 $\pm$ 6.3	41.63 $\pm$ 2.4	95.88 $\pm$ 2.6	85.10 $\pm$ 4.2	36.63 $\pm$ 3.7	109.89 $\pm$ 16.6	40.75 $\pm$ 2.4	87
– random erasing	115.08 $\pm$ 8.1	99.68 $\pm$ 7.1	48.93 $\pm$ 3.3	100.11 $\pm$ 5.9	88.79 $\pm$ 6.8	39.38 $\pm$ 3.1	127.81 $\pm$ 9.1	48.91 $\pm$ 3.7	87
+ horizontal flip	101.03 $\pm$ 4.1	86.79 $\pm$ 1.4	43.66 $\pm$ 0.9	103.78 $\pm$ 8.5	88.69 $\pm$ 8.3	39.90 $\pm$ 5.0	110.46 $\pm$ 2.0	45.31 $\pm$ 4.3	87